

41107 Marine and Ocean Engineering

Vortex-Induced Vibrations (VIV)

Solved MCQ Question Bank - professional formula notation

Scope and ordering

This PDF collects all reviewed questions beginning with **VORTEX INDUCED VIBRATIONS.** or **VIV.** Exact duplicates across the reviewed MCQ documents were removed. The questions are grouped by subtopic so the block is easier to study.

For each question, the structure is:

1. Original question statement and original answer options.
2. Correct answer highlighted immediately after the options.
3. **Formulario / principles used** with professional mathematical notation.
4. **Step-by-step solution** with numerical substitution whenever numerical data are present.

Default constants and notation

$$\rho = 1025 \text{ kg/m}^3, \quad \nu = 1.0 \times 10^{-6} \text{ m}^2/\text{s}, \quad A_c = \frac{\pi D^2}{4}$$

$$\omega = 2\pi f, \quad f = \frac{\omega}{2\pi}, \quad T = \frac{1}{f} = \frac{2\pi}{\omega}$$

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Natural frequency with added mass

N1. Question 1

VORTEX INDUCED VIBRATIONS. Calculate the un-damped natural frequency of a flexibly mounted circular cylinder in water where the following is given: the cylinder diameter $D = 0.2$ m, the mass (per unit length) $m = 4$ kg/m, the spring constant (per unit length) $k = 900$ kg/m and it is known that the normalized cross-flow vibration amplitude $A/D < 0.8$.

Options:

A. $f_n = 0.58$ Hz

B. $f_n = 2.39$ Hz

C. $f_n = 0.79$ Hz

Correct answer: C. $f_n = 0.79$ Hz

Note on the original stem. The original text writes $k = 900$ kg/m, but the calculation uses it as a stiffness per unit length, i.e. $k = 900$ N/m per metre span.

Formulario / principles used

- Cylinder cross-sectional area per unit length:

$$A_c = \frac{\pi D^2}{4}$$

- Added mass per unit length:

$$m' = \rho C_m A_c$$

For a circular cylinder in the potential-flow / small-amplitude estimate, use $C_m \approx 1$.

- Effective mass per unit length:

$$m_{\text{tot}} = m + m'$$

- Undamped angular natural frequency:

$$\omega_n = \sqrt{\frac{k}{m_{\text{tot}}}}$$

- Natural frequency in hertz:

$$f_n = \frac{\omega_n}{2\pi}$$

Step-by-step solution

1. Compute the cylinder area:

$$A_c = \frac{\pi D^2}{4} = \frac{\pi(0.2)^2}{4} = 0.0314 \text{ m}^2.$$

2. Compute the added mass per unit length:

$$m' = \rho C_m A_c = (1025)(1)(0.0314) = 32.2 \text{ kg/m}.$$

3. Compute the total effective mass:

$$m_{\text{tot}} = m + m' = 4.0 + 32.2 = 36.2 \text{ kg/m}.$$

4. Compute the angular natural frequency:

$$\omega_n = \sqrt{\frac{k}{m_{\text{tot}}}} = \sqrt{\frac{900}{36.2}} = \sqrt{24.86} = 4.99 \text{ rad/s}.$$

5. Convert from angular frequency to hertz:

$$f_n = \frac{\omega_n}{2\pi} = \frac{4.99}{2\pi} = 0.794 \text{ Hz} \approx 0.79 \text{ Hz}.$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test_Exam_all / 41107_MCQ_Study_Guide

N2. Question 2

VORTEX INDUCED VIBRATIONS. Calculate the undamped natural frequency of a flexibly mounted circular cylinder in water. Take $D = 0.08 \text{ m}$, structural mass per unit length $m = 4.0 \text{ kg/m}$, spring constant per unit length $k = 900 \text{ N/m}$, and assume $A/D < 0.8$ so that $C_m = 1$ for the added mass estimate. What is f_n approximately?

Options:

- A. 0.79 Hz
- B. 1.58 Hz
- C. 2.25 Hz
- D. 3.16 Hz

Correct answer: B. 1.58 Hz

Formulario / principles used

$$A_c = \frac{\pi D^2}{4}, \quad m' = \rho C_m A_c, \quad m_{\text{tot}} = m + m'$$

$$\omega_n = \sqrt{\frac{k}{m_{\text{tot}}}}, \quad f_n = \frac{\omega_n}{2\pi}$$

Step-by-step solution

1. Compute the cross-sectional area:

$$A_c = \frac{\pi(0.08)^2}{4} = 0.00503 \text{ m}^2.$$

2. Compute added mass per unit length:

$$m' = \rho C_m A_c = (1025)(1)(0.00503) = 5.15 \text{ kg/m}.$$

3. Compute total effective mass:

$$m_{\text{tot}} = m + m' = 4.0 + 5.15 = 9.15 \text{ kg/m}.$$

4. Compute the angular natural frequency:

$$\omega_n = \sqrt{\frac{900}{9.15}} = \sqrt{98.36} = 9.92 \text{ rad/s}.$$

5. Convert to hertz:

$$f_n = \frac{9.92}{2\pi} = 1.58 \text{ Hz}.$$

Source(s): 41107_mock_exam_with_answers

Free decay damping

D1. Question 3

VORTEX INDUCED VIBRATIONS. The amplitude response as function of time in a free decay test in water is shown in the following figure. What is the total damping factor (ζ) and the un-damped natural angular frequency (ω_n) when $y_n = 2$, $y_{n+1} = 0.5$ and $T = 1$ s?

Options:

- A. $\zeta = 0.096$; $\omega_n = 6.25$ rad/s
- B. $\zeta = 0.318$; $\omega_n = 5.96$ rad/s
- C. $\zeta = 0.221$; $\omega_n = 6.13$ rad/s

Correct answer: C. $\zeta = 0.221$; $\omega_n = 6.13$ rad/s

Formulario / principles used

- Logarithmic decrement from two successive peaks:

$$\delta = \ln \left(\frac{y_n}{y_{n+1}} \right)$$

- Course shortcut for damping factor:

$$\zeta \approx \frac{\delta}{2\pi}$$

- Damped angular frequency from the measured period:

$$\omega_d = \frac{2\pi}{T}$$

- Standard exact relation:

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \quad \Rightarrow \quad \omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}}$$

Step-by-step solution

1. Compute the logarithmic decrement:

$$\delta = \ln \left(\frac{y_n}{y_{n+1}} \right) = \ln \left(\frac{2}{0.5} \right) = \ln(4) = 1.386.$$

2. Compute the damping factor using the course shortcut:

$$\zeta \approx \frac{\delta}{2\pi} = \frac{1.386}{2\pi} = 0.221.$$

3. Compute the damped angular frequency from the period:

$$\omega_d = \frac{2\pi}{T} = \frac{2\pi}{1} = 6.28 \text{ rad/s.}$$

4. Compare with the answer options. Only option C contains the correct damping factor:

$$\zeta = 0.221.$$

5. Therefore the intended MCQ answer is option C.

Important note. With the strict standard vibration relation, $\omega_n = \omega_d / \sqrt{1 - \zeta^2}$, the numerical value would be $6.28 / \sqrt{1 - 0.221^2} = 6.44$ rad/s. Therefore, the frequency value in the original option appears to follow the answer-set/course convention rather than the strict exact relation. For the MCQ, select option C.

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test_Exam_all / 41107_MCQ_Study_Guide

D2. Question 4

VORTEX INDUCED VIBRATIONS. The amplitude response in a free-decay test is shown in the figure. If $y_n = 1.5$, $y_{n+1} = 0.9$ and $T = 0.8$ s, what are the damping ratio ζ and the undamped natural angular frequency ω_n approximately?

Options:

- A. $\zeta = 0.081$, $\omega_n = 7.88$ rad/s
- B. $\zeta = 0.081$, $\omega_n = 6.28$ rad/s
- C. $\zeta = 0.163$, $\omega_n = 7.88$ rad/s
- D. $\zeta = 0.163$, $\omega_n = 6.28$ rad/s

Correct answer: A. $\zeta = 0.081$, $\omega_n = 7.88$ rad/s

Formulario / principles used

$$\delta = \ln \left(\frac{y_n}{y_{n+1}} \right), \quad \zeta \approx \frac{\delta}{2\pi}$$

$$\omega_d = \frac{2\pi}{T}, \quad \omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}}$$

Step-by-step solution

1. Compute logarithmic decrement:

$$\delta = \ln \left(\frac{1.5}{0.9} \right) = \ln(1.6667) = 0.5108.$$

2. Estimate damping ratio:

$$\zeta \approx \frac{0.5108}{2\pi} = 0.0813 \approx 0.081.$$

3. Compute damped angular frequency:

$$\omega_d = \frac{2\pi}{0.8} = 7.854 \text{ rad/s.}$$

4. Correct to undamped angular natural frequency:

$$\omega_n = \frac{7.854}{\sqrt{1 - 0.0813^2}} = 7.88 \text{ rad/s.}$$

5. Therefore the correct option is A.

Source(s): 41107_mock_exam_with_answers

Reduced velocity

R1. Question 5

VORTEX INDUCED VIBRATIONS. Plane oscillatory flow around a free, smooth, flexibly-mounted, circular cylinder. What is the reduced velocity when the oscillatory flow is defined by $U[\text{m/s}] = 2.5 \cdot \sin(\omega t)$, $D = 0.5 \text{ m}$, and the angular natural frequency $\omega_n = 7.85 \text{ rad/s}$?

Options:

- A. $V_r = 6.25$
- B. $V_r = 0.25$
- C. $V_r = 4.0$

Correct answer: C. $V_r = 4.0$

Formulario / principles used

- Natural frequency in hertz:

$$f_n = \frac{\omega_n}{2\pi}$$

- Reduced velocity in oscillatory flow:

$$V_r = \frac{U_m}{Df_n}$$

where U_m is the maximum oscillatory velocity, not the mean velocity.

Step-by-step solution

- Read the velocity amplitude from $U(t)$:

$$U(t) = 2.5 \sin(\omega t) \Rightarrow U_m = 2.5 \text{ m/s.}$$

- Convert angular natural frequency to ordinary frequency:

$$f_n = \frac{\omega_n}{2\pi} = \frac{7.85}{2\pi} = 1.25 \text{ Hz.}$$

- Compute the natural-frequency velocity scale:

$$Df_n = (0.5)(1.25) = 0.625 \text{ m/s.}$$

- Compute reduced velocity:

$$V_r = \frac{U_m}{Df_n} = \frac{2.5}{0.625} = 4.0.$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test_Exam_all / 41107_MCQ_Study_Guide

R2. Question 6

VORTEX INDUCED VIBRATIONS. Plane oscillatory flow around a free, smooth, flexibly-mounted, circular cylinder. What is the reduced velocity when the oscillatory flow is defined by $U[\text{m/s}] = 0.5 \cdot \sin(\omega t)$, $D = 1 \text{ m}$, and the angular natural frequency $\omega_n = 15.71 \text{ rad/s}$?

Options:

- A. $V_r = 0.8$

B. $V_r = 1.25$

C. $V_r = 0.2$

Correct answer: C. $V_r = 0.2$

Formulario / principles used

$$f_n = \frac{\omega_n}{2\pi}, \quad V_r = \frac{U_m}{Df_n}$$

Step-by-step solution

1. Extract the maximum velocity:

$$U(t) = 0.5 \sin(\omega t) \Rightarrow U_m = 0.5 \text{ m/s.}$$

2. Convert angular natural frequency:

$$f_n = \frac{15.71}{2\pi} = 2.50 \text{ Hz.}$$

3. Compute the velocity scale:

$$Df_n = (1)(2.50) = 2.50 \text{ m/s.}$$

4. Compute the reduced velocity:

$$V_r = \frac{0.5}{2.50} = 0.20.$$

Source(s): Test MC exam_Corr / Test_Exam_all

R3. Question 7

VORTEX INDUCED VIBRATIONS. Plane oscillatory flow around a free, smooth, flexibly-mounted circular cylinder. The oscillatory flow is $U(t) = 1.8 \sin(\omega t)$ m/s, the diameter is $D = 0.40$ m, and the angular natural frequency is $\omega_n = 9.42$ rad/s. What is the reduced velocity V_r ?

Options:

A. $V_r = 2.0$

B. $V_r = 3.0$

C. $V_r = 4.5$

D. $V_r = 6.0$

Correct answer: B. $V_r = 3.0$

Formulario / principles used

$$f_n = \frac{\omega_n}{2\pi}, \quad V_r = \frac{U_m}{Df_n}$$

Step-by-step solution

1. Extract the maximum velocity:

$$U(t) = 1.8 \sin(\omega t) \quad \Rightarrow \quad U_m = 1.8 \text{ m/s.}$$

2. Convert angular frequency to hertz:

$$f_n = \frac{9.42}{2\pi} = 1.50 \text{ Hz.}$$

3. Compute Df_n :

$$Df_n = (0.40)(1.50) = 0.60 \text{ m/s.}$$

4. Compute reduced velocity:

$$V_r = \frac{1.8}{0.60} = 3.0.$$

Source(s): 41107_mock_exam_with_answers

Amplitude ratios and vibration direction

A1. Question 8

VORTEX INDUCED VIBRATIONS. Steady current around a free, flexibly-mounted circular cylinder. What is the ratio between the maximum in-line vibration amplitude and the maximum cross-flow vibration amplitude, A_x/A_y ? (values below 1 indicate cross-flow vibration amplitude is larger than in-line vibration amplitude)?

Options:

- A. Approximately 1/10
- B. Approximately 10
- C. Approximately 1

Correct answer: A. Approximately 1/10

Formulario / principles used

- Typical Vortex-Induced Vibrations (VIV) amplitude relation:

$$\frac{A_x}{A_y} \approx 0.1 = \frac{1}{10}.$$

- A_x : in-line vibration amplitude, mainly related to oscillating drag.
- A_y : cross-flow vibration amplitude, mainly related to oscillating lift.

Step-by-step solution

1. Identify the requested ratio:

$$\frac{A_x}{A_y} = \frac{\text{in-line amplitude}}{\text{cross-flow amplitude}}.$$

2. In typical VIV, cross-flow vibration is much larger than in-line vibration:

$$A_y \approx 10A_x.$$

3. Rearrange to the requested ratio:

$$\frac{A_x}{A_y} \approx \frac{A_x}{10A_x} = \frac{1}{10} = 0.1.$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test_Exam_all / 41107_MCQ_Study_Guide

A2. Question 9

VORTEX INDUCED VIBRATIONS. Steady current around a free, flexibly-mounted circular cylinder. What is the ratio between the maximum in-line vibration amplitude and the maximum cross-flow vibration amplitude, A_x/A_y ? (values below 1 indicate cross-flow vibration amplitude is larger than in-line vibration amplitude)?

Options:

- A. approximately 1/10
- B. approximately 1

C. approximately 1/100

Correct answer: A. approximately 1/10

Formulario / principles used

$$\frac{A_x}{A_y} \approx 0.1 = \frac{1}{10}$$

Values below 1 mean that $A_y > A_x$, i.e. cross-flow amplitude is larger.

Step-by-step solution

1. The question asks for A_x/A_y , not A_y/A_x .

2. Typical VIV gives:

$$A_x \approx 0.1A_y.$$

3. Substitute directly into the ratio:

$$\frac{A_x}{A_y} \approx \frac{0.1A_y}{A_y} = 0.1 = \frac{1}{10}.$$

4. Therefore option A is correct.

Source(s): Test MC exam_Corr / Test_Exam_all

Stability parameter and amplitude response

S1. Question 10

VORTEX INDUCED VIBRATIONS. Steady current around a free, flexibly-mounted, smooth, circular cylinder. The mass ratio $M = 5$, and the structural damping factor $\zeta_s = 0.05$. What is the expected cross-flow amplitude response assuming the following model for the amplitude response as function of stability parameter: $A/D = 1.5/K_s$?

Options:

A. $A/D = 0.72$

B. $A/D = 0.96$

C. $A/D = 0.48$

Correct answer: C. $A/D = 0.48$

Formulario / principles used

- Structural logarithmic decrement:

$$\delta_s = 2\pi\zeta_s$$

- Stability parameter:

$$K_s = 2M\delta_s = 4\pi M\zeta_s$$

- Given empirical amplitude model:

$$\frac{A}{D} = \frac{1.5}{K_s}$$

Step-by-step solution

1. Compute the logarithmic decrement:

$$\delta_s = 2\pi\zeta_s = 2\pi(0.05) = 0.314.$$

2. Compute the stability parameter:

$$K_s = 2M\delta_s = 2(5)(0.314) = 3.14.$$

3. Use the amplitude model from the question:

$$\frac{A}{D} = \frac{1.5}{K_s} = \frac{1.5}{3.14} = 0.477.$$

4. Round to the closest option:

$$\frac{A}{D} \approx 0.48.$$

Source(s): Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide

S2. Question 11

VIV - Stability Parameter and Suppression. A marine riser with the following properties: $D = 0.3$ m, structural mass per unit length $m = 50$ kg/m, structural damping $\zeta_s = 0.01$. Will VIV be a concern?

Options:

- A.** No - the stability parameter is large enough to suppress VIV
- B.** Yes - the stability parameter is small, significant VIV is expected
- C.** Cannot determine without current velocity

Correct answer: B. Yes - the stability parameter is small, significant VIV is expected

Formulario / principles used

- Displaced-fluid mass per unit length using the source's simplified convention:

$$m_d = \rho A_c = \rho \frac{\pi D^2}{4}$$

- Mass ratio:

$$M = \frac{m}{m_d}$$

- Structural logarithmic decrement and stability parameter:

$$\delta_s = 2\pi\zeta_s, \quad K_s = 2M\delta_s$$

- Interpretation: small K_s means low damping/light structure and higher VIV susceptibility.

Step-by-step solution

1. Compute the displaced-fluid mass per unit length:

$$m_d = \rho \frac{\pi D^2}{4} = 1025 \frac{\pi (0.3)^2}{4} = 72.4 \text{ kg/m.}$$

2. Compute the mass ratio:

$$M = \frac{m}{m_d} = \frac{50}{72.4} = 0.69.$$

3. Compute structural logarithmic decrement:

$$\delta_s = 2\pi\zeta_s = 2\pi(0.01) = 0.0628.$$

4. Compute stability parameter:

$$K_s = 2M\delta_s = 2(0.69)(0.0628) = 0.087.$$

5. Since K_s is very small, the riser is not strongly damped or heavy enough to suppress VIV.

6. Therefore significant VIV is expected.

Source(s): 41107_MCQ_Study_Guide

In-line instability regions and lock-in

I1. Question 12

VORTEX INDUCED VIBRATIONS. Steady current around a free, flexibly-mounted circular cylinder. Which in-line instability region (vibration) is associated with normal vortex shedding when $St = 0.2$?

Options:

- A. 2nd instability region ($V_r \approx 2.5$)
- B. 3rd instability region ($V_r \approx 5$)
- C. 1st instability region ($V_r \approx 1.7$)

Correct answer: C. 1st instability region ($V_r \approx 1.7$)

Formulario / principles used

- Strouhal relation:

$$St = \frac{f_v D}{U}$$

- Reduced velocity:

$$V_r = \frac{U}{D f_n}$$

- Cross-flow lock-in scale:

$$V_r \approx \frac{1}{St}$$

For $St = 0.2$, this gives $V_r \approx 5$, which is the cross-flow lock-in scale, not the requested in-line instability region.

- Course MCQ convention: the in-line instability associated with normal vortex shedding is the 1st in-line instability region around $V_r \approx 1.7$.

Step-by-step solution

1. The question specifically asks for an **in-line** instability region, not the cross-flow lock-in region.
2. Check the common cross-flow lock-in scale:

$$V_r \approx \frac{1}{St} = \frac{1}{0.2} = 5.$$

3. This value corresponds to the cross-flow lock-in scale, so option B is the typical trap.
4. In the course convention, normal vortex shedding is associated with the 1st in-line instability region.
5. Therefore the correct option is:

1st instability region ($V_r \approx 1.7$).

Source(s): Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide

Final formula reminder for this group

Natural frequency with added mass

$$A_c = \frac{\pi D^2}{4}$$

$$m' = \rho C_m A_c$$

$$m_{\text{tot}} = m + m'$$

$$\omega_n = \sqrt{\frac{k}{m_{\text{tot}}}}, \quad f_n = \frac{\omega_n}{2\pi}$$

Free decay damping

$$\delta = \ln \left(\frac{y_n}{y_{n+1}} \right)$$

$$\zeta \approx \frac{\delta}{2\pi}$$

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}} \quad (\text{exact logarithmic decrement relation})$$

$$\omega_d = \frac{2\pi}{T_d}, \quad \omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}}$$

Reduced velocity and lock-in

$$f_n = \frac{\omega_n}{2\pi}$$

$$V_r = \frac{U}{Df_n} \quad (\text{steady current}), \quad V_r = \frac{U_m}{Df_n} \quad (\text{oscillatory flow})$$

$$St = \frac{f_v D}{U}$$

Cross-flow lock-in scale: $V_r \approx \frac{1}{St}$

Amplitude response and stability parameter

$$\delta_s = 2\pi\zeta_s$$

$$K_s = 2M\delta_s = 4\pi M\zeta_s$$

$$\frac{A}{D} = \frac{1.5}{K_s} \quad \text{when this empirical model is given}$$

$$\frac{A_x}{A_y} \approx 0.1 = \frac{1}{10}$$